

School of Electrical and Computer Engineering

Department of control system Engineering

Introduction to control system

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Chapter one
Introduction to control system

Introduction

- Automatic control has played a vital roll in the advanced of engineering and science. In addition to the extreme importance in space-vehicle system, missile guidance system, robotics and so on..
- Advance in the theory and practice of automatic control provide the means for attaining
    - Optimum performance of dynamic systems
    - Improving productivity
    - To Avoid many routine repetitive manual operations

Todays control Engineering has application in almost every area of technology.

- **Industrial process control**

  Industrial process control covers the control of temperature, pressure, flow-level and any other industrial applications

- **Drive control (speed control)**

  This group includes speed control of motors on different machines such as in plastic manufacturing, paper production or textile machines. specially designed controllers are normally used for these application since they have to remain stable during fast disturbance in the range of tenths of seconds

- **Control of electrical variable**

  This refers to stabilization of electrical parameters, e.g. . voltage, current, power or even frequency

- **Position control**

  Involves the position of tools, work pieces or complete assemblies either in two or three dimensions. e.g. antenna, positioning of guns and tanks, drone, CNC machines

- **Course control**

  Refers the control of the course of ship or plane (Autopilots) here the controller has to satisfy special demands, such as high processing speed and operational safety, combined with weight.

Basic Definitions

- **System:** is an arrangement or combination of different physical components that are connected or related together to form an entire unit to achieve a certain objective.
- **Control:** the meaning of control is regulate, direct or command a system to achieve a desired objective.
- **Plant:** it is defined as the portion of a system in which is to be control or regulate, it is also called process.
- **Controller:** it is an element within the system itself, or external to the system and it control the plant or the process.
- **Input:** the applied signal or excitation to the control system.
- **Output:** the actual response obtained from the control system.

- **Model:** A conceptual model representation of a system, made of the composition of concepts which are used to help people know, understand, or simulate a subject the model represents.
- **Mathematical model:** is a description of a system using mathematical concepts and representation of a certain system or part of system.
- **Disturbance or noise**: is a signal which enters the system at a point other than the reference input and has the effect of undermining the normal system operation.

Control system

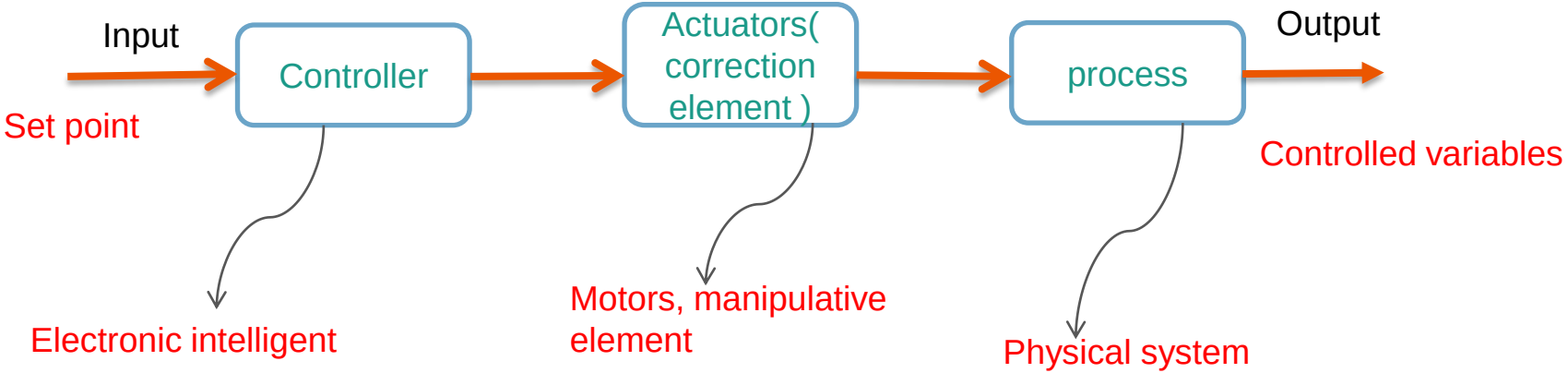
- A control system is an interconnection of **components forming** a system configuration that will provide a **desired system response**. *i.e.* a control system is by which any quantity of interest is measured or altered in accordance with a desired manner.
- The basis for analysis of a system is the foundation provided by **linear system theory**, which assumes a **cause-effect** relationship for the component of a system.

- Therefore a component or process to be controlled can be represented by block **showing input-output** relationships. **Cause and effect** of relationship of the process



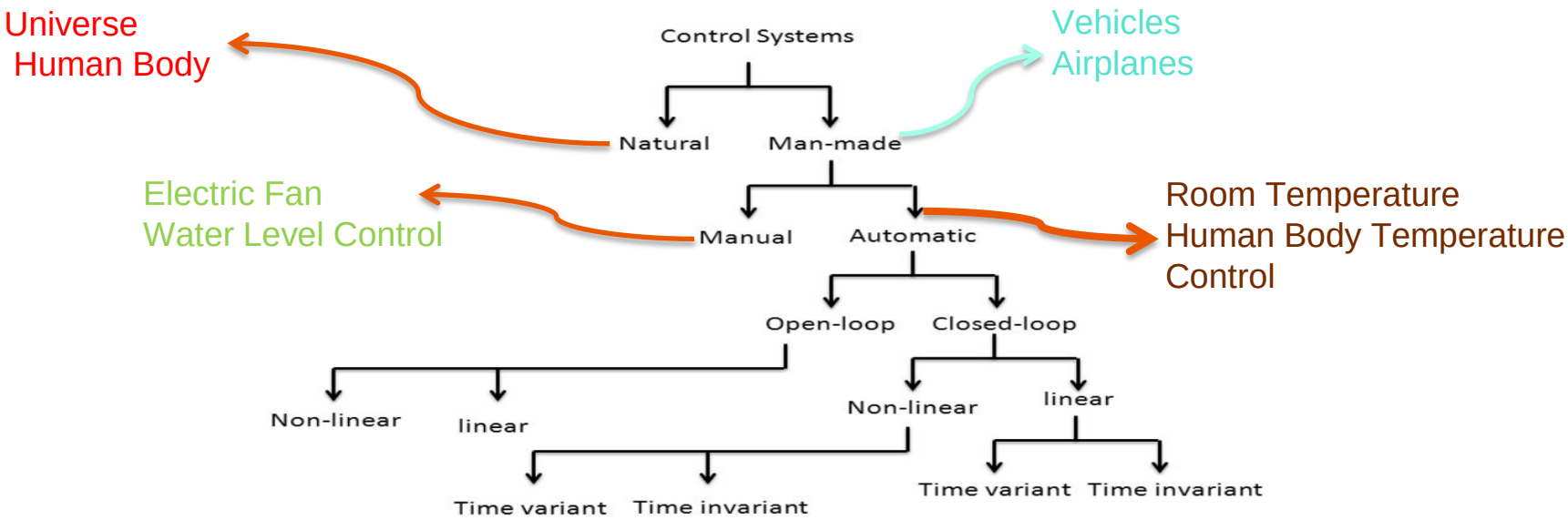
- Is a system where the output of the system is controlled to be at some **specific value** or to change in some **prescribed** way as determined by the input to the system.
- A control System is a device, or set of devices to manage, command, direct or regulate the behavior of other device(s) or system(s)

- Every controller has (at least) a controller and an actuator(also called a final control element)



A block diagram for a control system

Classifications of control systems

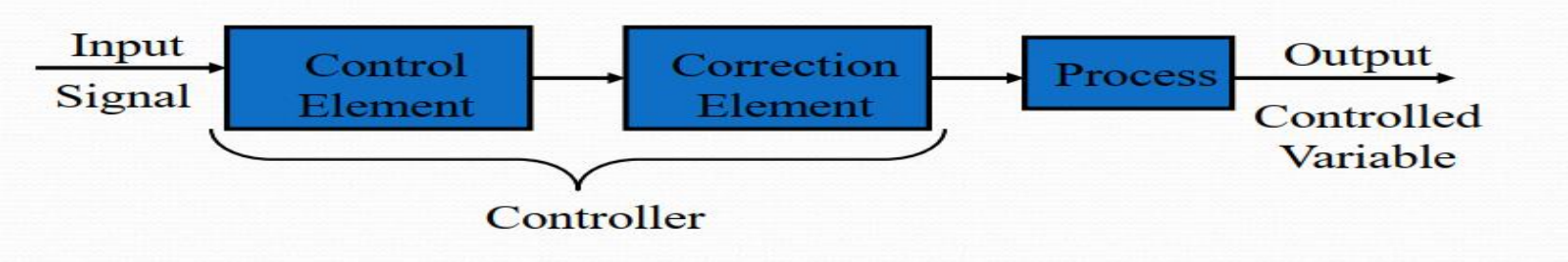


Types of control system

- There are two major configuration of control systems (internal architecture of the total system)
    - Open loop control system
    - Closed loop (feedback system)

Open loop

- System which control action **doesn't depend** on out put is called **open loop** system automatically correct for variations in its output
- No information is feedback to the system to adjust itself and maintain a constant output
- An input is chosen on the basis of experience of such a system to give the value of the required output.



Examples

- An **electric heater** with a selection switch
- Systems which operate by preset timing mechanisms like **Traffic Lights**, **Washing Machine**

Advantages

- Relatively simple hence low cost with generally good reliability
- Inherently Stable
- Easy maintenance

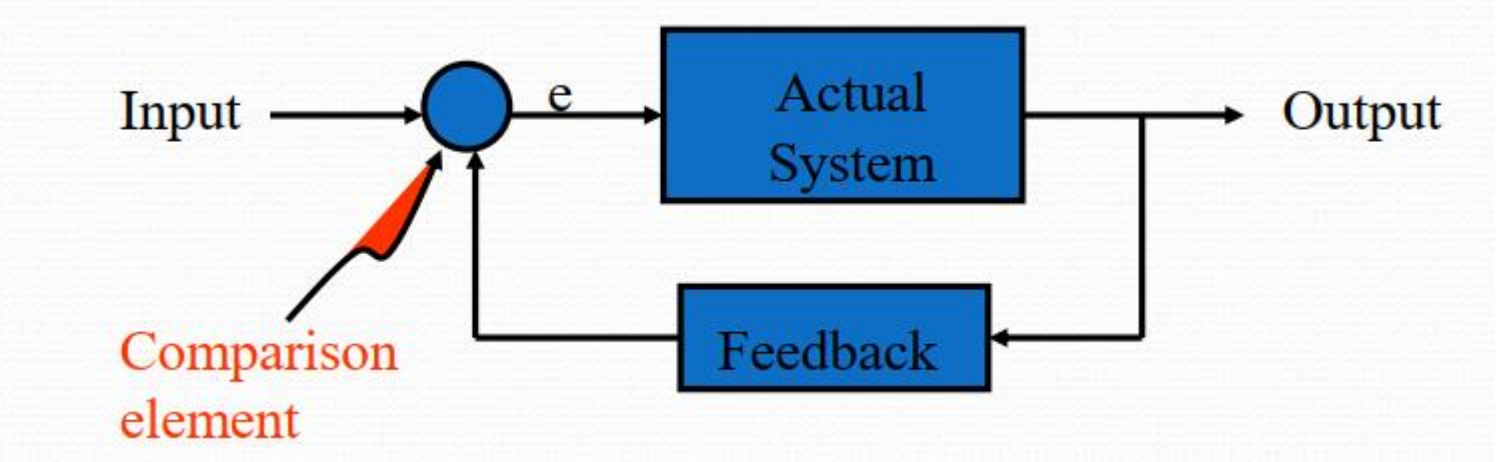
Disadvantages

- Often inaccurate since there is no error correction
- More sensitive to changes in component characteristics
- More sensitive to disturbances
- Limitation on applications

Closed loop

- If control actions of the system are somehow **dependent on** output or change in out put the system is called **closed loop system**.
- In contrast to an open loop control system utilizes an additional measure of the actual output to compare the actual output with the desired output responses.
- The measure of the output is called the **feedback signal**.
- **A feedback control** system is a control system that tends to maintain a **prescribed** relationship of one system variables to another by comparing function of these variables and using the difference as a **means of control**.

- A feedback signal to the input from the output will be sent and used to modify the input so that the output is maintained constant regardless of any conditions



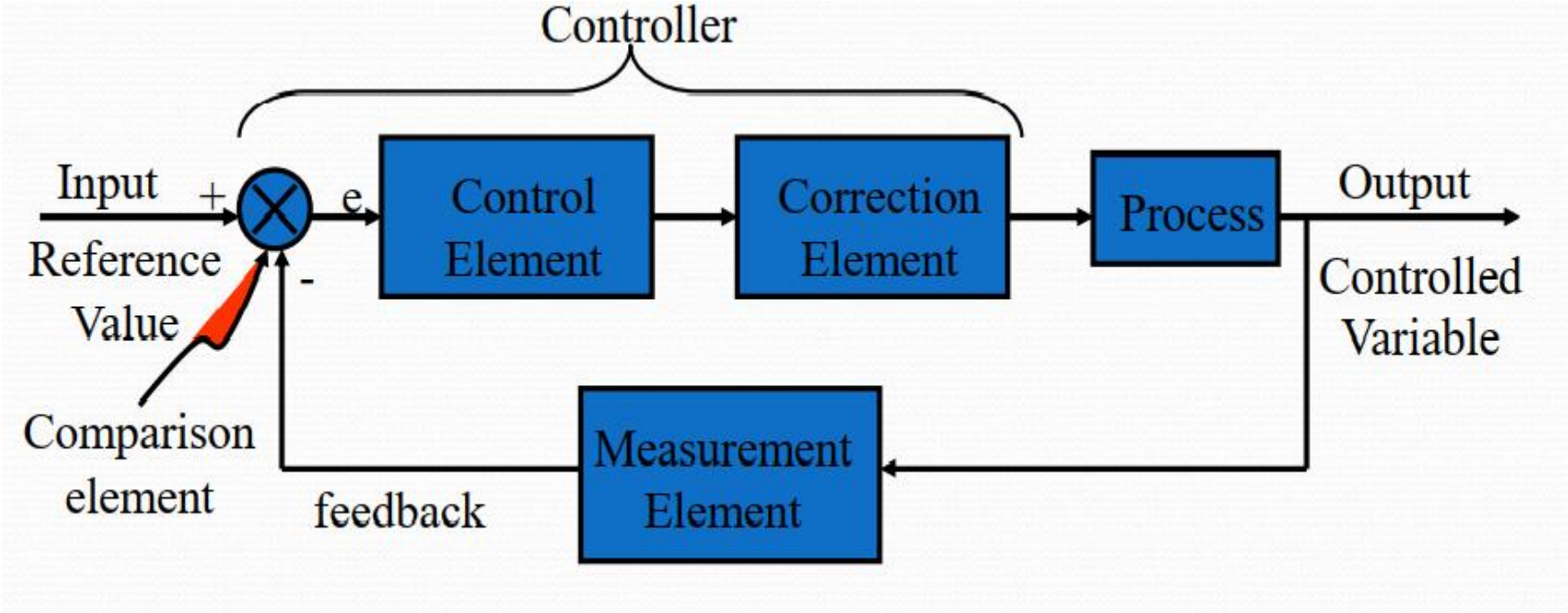
feedback control system

Basic Components

- Any Control System consists of a number of basic subsystems or elements
- Those elements could appear separate or integrated as a single entity

The basic elements are:

- Comparison Element
- Control Element
- Correction Element
- Process Element (plant)
- Measurement Element (Transducers)


Controller
Input
Reference
Value
+
-
e
Control
Element
Correction
Element
Process
Output
Controlled
Variable
Comparison
element
feedback
Measurement
Element

1. ***Comparison Element:*** compares the reference value of the variable being controlled with the measured value of the actual output and produces an error signal.
2. ***Control Element:*** decides what action to be taken after it gets an error signal.
3. ***Correction Element (actuator):*** used to produce a change in the process in order to avoid the error.
4. ***Process Element (Plant):*** is the system of which a variable is being controlled
5. ***Measurement Element:*** Produces a Signal related to the actual output and provides a feedback signal to the comparison element.

Advantages

- Ability to match the actual to the required value since there is continuous error correction
- Less sensitivity to disturbances
- Increased speed of response and hence increased band width
- Increase accuracy
- Reduce the effect of non linearity

Disadvantages

- These systems are complicated in design and costlier
- These system may become unstable, because of time delays when transferring corrective action

Importance of feedback

The presence of feedback imports the following properties to the system

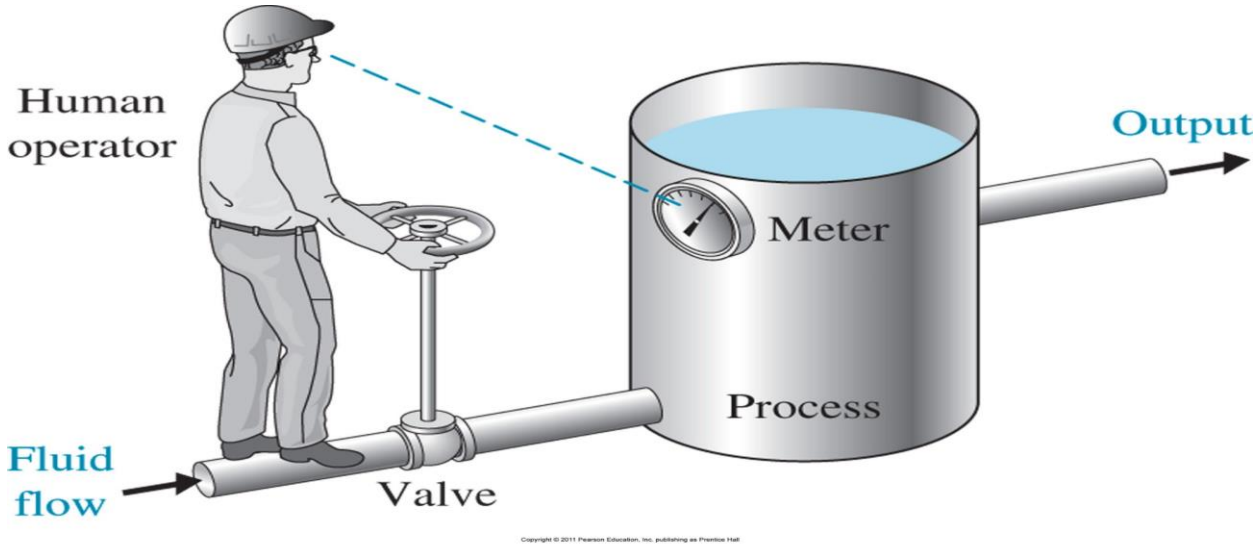
- **Increase accuracy:** for example ability to faithfully reproduce the input
- Tendency towards **oscillation** or **instability**
- **Reduced sensitivity** of the ratio of output to input to variation in the system parameters and other characteristic.
- Reduced effect of **external disturbance** or noise
- Increase bandwidth

The most important tasks of control Engineers

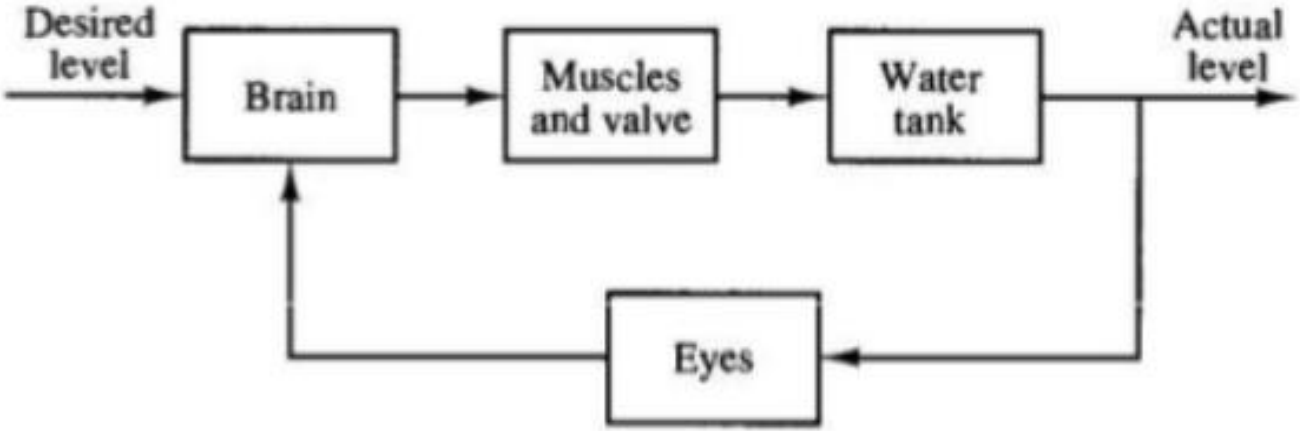
- Determine the process variable
- Checking whether automatic control offers significant advantages
- Determining the measurement site
- Selecting the manipulator
- Selecting disturbance
- Selecting a suitable controller
- Installation of the controller in accordance with applicable regulation
- Starting up adjusting parameters, optimizing an unsatisfactory control loop

Examples

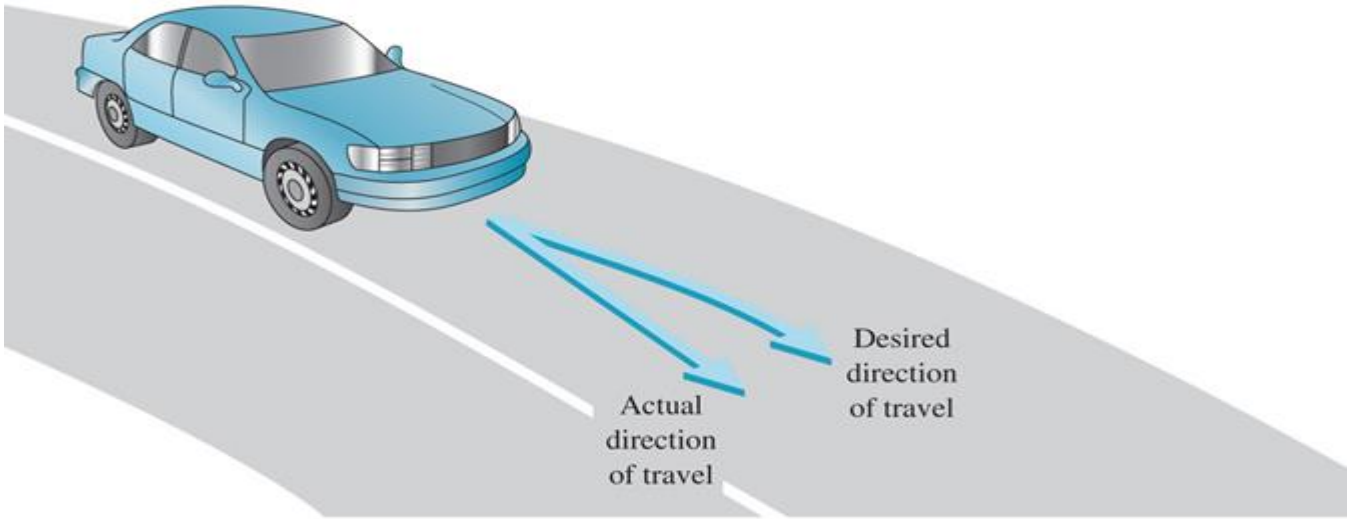
1. A manual control system for regulating the level of fluid in a tank by adjusting the output valve. The operator views the level of fluid through a port in the side of the tank

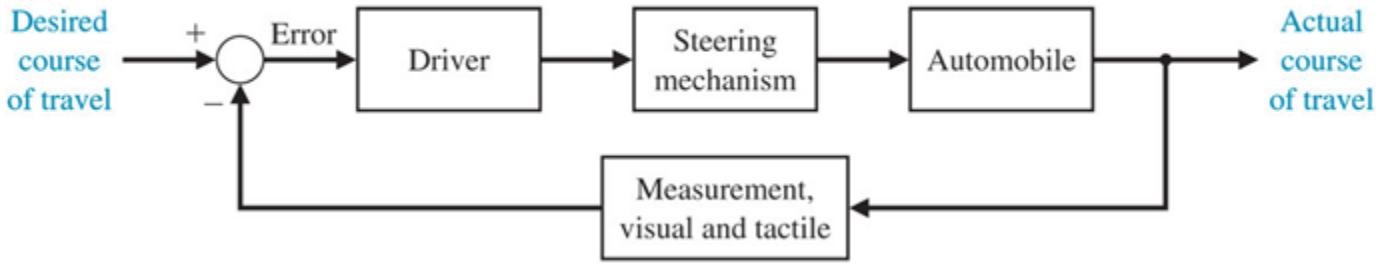


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Desired level
Brain
Muscles and valve
Water tank
Actual level
Eyes

2. The driver uses the difference between the actual and the desired direction of travel to generate a controlled adjustment of the steering wheel.




Desired course of travel
+
−
Error
Driver
Steering mechanism
Automobile
Actual course of travel
Measurement, visual and tactile

Main control strategies

Every control system must guarantee first the stability of the closed-loop behavior. The possibility to fulfill different specifications varies model considered and the control system chosen.

Classical controls

- Control methods where the ODE that describe a system are transformed using Laplace, Fourier, or Z-transforms and manipulated in the transform domain.

Modern control

- Methods where high-order differential equations are broken into a system of first-order equations. The input, output and internal states of the system are described by vector called “state variables”

Adaptive control

- Adaptive control uses on-line identification of the process parameters, or modification of controller gain, there by obtaining strong robustness properties. The controller changes it is response characteristics over time to better control the system

Robust control

- Robust control deals explicitly with uncertainty in its approach to controller design.

Optimal control

- In a system, performance matrices are identified and arranged into a cost function. The cost function is minimized to create an operational system with the lowest cost .

Nonlinear controls

- The youngest branch of control engineering it is used for the system that can not be described by linear equations or ODE

Intelligent control

- Intelligent control uses varies AI (artificial intelligent) computing techniques like Neural network, fuzzy logic, machine learning, evolutionary computation and genetic algorithms to control a dynamic system.

Stochastic control

- Deals with uncertainty in the model like random noise and disturbance must take into account.

Classical control Vs Modern control

Classical control theory

- It is the branch of control theory that deals with the behavior of dynamical systems with inputs and how their behavior is modified by feedback, using the Laplace transform as a basic tool to model such system.
- Classical Control Theory deal with LTIS-SISO system.
- Classical Control Theory uses the Laplace transform to change an ODE in the time domain into regular algebraic polynomial in the transform domain.

Modern Control Theory

- Instead of changing domains to avoid the complexities of time domain ODE mathematics, converts the differential equations into a system of lower order time domain equations called state equations, which can then be manipulated using techniques from liner algebra.
- State-space: representation is a mathematical model of a physical system as a set of input, output and state variables related by first-order differential equation or difference equations.
- State variables are variables whose values evolves through time in the way that depends on the values they have at any given time and also depends on the externally imposed values of input variables.

Review of Laplace Transform

- The method of transforming ***t domain to s*** domain is known as Laplace transform,

  Where s, is a complex operator denoted by $s = \sigma + j\omega$.

- In control system setting up the system equations, and there solution is required

$$L[f(t)] = \int_0^\infty f(t)e^{-st}dt$$

Example : Consider the following second-order differential equation with initial conditions taken to be zero.

$$f(t) = \frac{dx^2(t)}{dt^2} + 5\frac{dx(t)}{dt} + 10\,x(t)$$

Some standard Laplace Transform

Laplace Transform $F(s)$	Time Function $f(t)$
1	Unit-impulse function $\delta(t)$
$\frac{1}{s}$	Unit-step function $u_s(t)$
$\frac{1}{s^2}$	Unit-ramp function t
$\frac{n!}{s^{n+1}}$	$t^n (n = \text{positive integer})$
$\frac{1}{s+\alpha}$	$e^{-\alpha t}$
$\frac{1}{(s+\alpha)^2}$	$te^{-\alpha t}$
$\frac{n!}{(s+\alpha)^{n+1}}$	$t^n e^{-\alpha t} (n = \text{positive integer})$
$\frac{1}{(s+\alpha)(s+\beta)}$	$\frac{1}{\beta-\alpha}(e^{-\alpha t} - e^{-\beta t})(\alpha \neq \beta)$

Laplace transform s-domain (F(s))	Time(t)-domain (f(t))
$\frac{s}{(s+\alpha)(s+\beta)}$	$\frac{1}{\beta-\alpha}(\beta e^{-\beta t}-\alpha e^{-\alpha t})(\alpha \neq \beta)$
$\frac{1}{s(s+\alpha)}$	$\frac{1}{\alpha}(1-e^{-\alpha t})$
$\frac{1}{s(s+\alpha)^2}$	$\frac{1}{\alpha^2}(1-e^{-\alpha t}-\alpha t e^{-\alpha t})$
$\frac{1}{s^2(s+\alpha)}$	$\frac{1}{\alpha^2}(\alpha t-1+e^{-\alpha t})$
$\frac{1}{s^2(s+\alpha)^2}$	$\frac{1}{\alpha^2}\left[t-\frac{2}{\alpha}+\left(t+\frac{2}{\alpha}\right)e^{-\alpha t}\right]$
$\frac{s}{(s+\alpha)^2}$	$(1-\alpha t)e^{-\alpha t}$

Systems Classification

- Linear and non-linear system
- Causal and non -causal system
- Stable and unstable system
- Time variant and time invariant system

Linear system: A system is linear if it obeys principle of **superposition**.

1. Law of additive
2. Law of homogeneity

Causal system: the system is causal if the output of the system is **independent** of future value input.

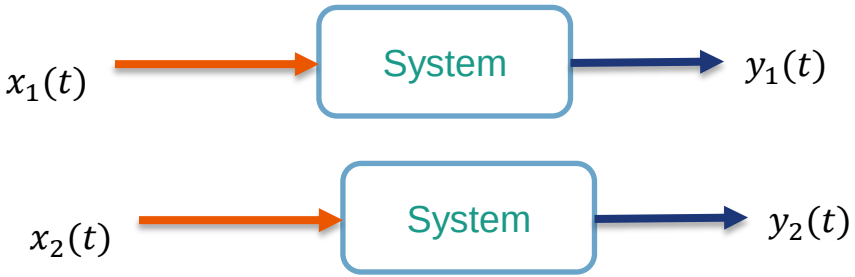
Stable: Bounded i/p – Bounded o/p criteria(BIBO), for a stable system o/p should be bounded for bounded i/p at each and every instant of time.

Time invariant: A system is said to be time variant if its input and output characteristics vary with time. Otherwise, the system is considered as time invariant.

Linear system

Given two pair of inputs and outputs, $y(t) = x\ (t)$

- $y_1(t) = x_1(t)$ and $y_1(t) = x_1(t)$



- $y(t) = x_1(t) + x_2(t)$



2. Homogeneity

